

ESF: Harmonizing Heterogeneous EEG Acquisition for Pretrained Classifiers

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Abstract—A pretrained EEG classifier is built on recordings from one setup and then used on recorders that sample, amplify, reference and filter differently. We ask a narrow question: if every recording is first passed through one fixed preprocessing contract, which of those differences stop mattering, for which model, and at what cost? The contract is the Encephlian Standard Format (ESF), six standard preprocessing steps in a fixed order. We simulate five kinds of recorder difference (sampling rate, gain, mains interference, missing or bipolar electrodes, broadband noise) on raw microvolt recordings of the TUAB evaluation set ($n=276$, held out from all trainings), and score three frozen public models (EEGPT, LaBraM, BIOT) with the contract complete and with the one step that addresses each difference switched off. Gain and mains differences are fully undone for every model. A low sampling rate (128 Hz) is undone for EEGPT but not for BIOT or LaBraM, and a low-pass sweep shows why: EEGPT ignores everything above 15 Hz, while the other two depend on the 64–100 Hz band that low-rate acquisition removes. Missing electrodes and broadband noise are undone for none. Without the contract, two models turn normal recordings into confident abnormal reports and the third turns abnormal recordings into confident normal ones; with it, those failures disappear exactly where the sweep says they should. Controls, a label-free monitor that fails, and six recordings from a 125 Hz Natus recorder complete the study. v2 corrects v1 (Version note). Code, probes and per-recording predictions are released.

Index Terms—EEG, canonical representation, acquisition shift, pretrained models, recoverability, selective prediction.

I. INTRODUCTION

Self-supervised EEG models [1–3] score well on the Temple University corpora [4, 5], which were recorded on one hospital’s equipment. A clinic that wants to use such a model has different equipment: another sampling rate, another gain convention, fewer electrodes, 50 rather than 60 Hz mains, often a legacy recorder with a proprietary export. The usual answer is a preprocessing step that maps every recording to one format. Whether that step actually restores the model’s accuracy, and for which differences, is rarely measured; existing robustness benchmarks corrupt the model input directly rather than simulating the recorder.

This paper measures it (Fig. 1). The format is the **Encephlian Standard Format (ESF)**: six standard preprocessing steps, fixed in order and applied identically at training and deployment; we call it the contract. For each of five recorder differences we build a physical simulation on the raw microvolt signal, then run the same frozen model twice: once with the full contract (the *canonicalized* arm) and once with the single contract step that addresses that difference

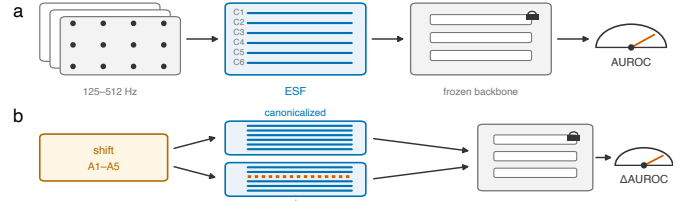


Fig. 1: **Deployment and protocol.** (a) Recorders that differ in rate, channel set, gain, mains and export feed one deterministic six-step contract (ESF) and one frozen model and classifier; nothing downstream is retrained. (b) An acquisition operator acts on the raw microvolt recording; the canonicalized arm applies the full contract, the naive arm the same contract with the one stage that addresses the axis switched off; both feed the same frozen model, so their difference is the recovery attributable to that step.

switched off (the *naive* arm). The gap between the two arms is what that step is worth. Because the recordings, the model weights and the classifier on top are identical in both arms, the comparison is exact. Contributions: (i) the protocol and its simulators, released; (ii) a short theory that says in advance which differences a deterministic step can undo, and which depend on the model; (iii) three frozen models, each with four classifiers, so every number carries a spread; (iv) a per-model envelope (Table 1) stating which recorder settings each model tolerates, and a bandwidth measurement that explains the one setting on which the models disagree; (v) controls, a label-free monitor that fails, and a real legacy recorder.

II. BACKGROUND

BIOT tokenizes heterogeneous biosignals [3], LaBraM scales masked-token pretraining to 2,500 h [2], EEGPT adds representation alignment [1]; linear probing of a frozen backbone is the standard evaluation [6]. The in-corpus to deployment gap defines clinical machine learning [7–9]; ImageNet-C [10] set the parameterized perturbation methodology adapted here. BIDS-EEG [11] and NWB [12] standardize datasets; ESF standardizes the signal a model sees. Three contemporaneous lines work from the model side: BRIDGE-EEG [13] maps recordings to a device-agnostic 62-channel time–frequency representation before pretraining its own encoders; PRISM [14] pretrains across acquisition systems; graph montage harmonization [15] learns a mapping between electrode layouts. None simulates the recorder as operators,

holds a fixed contract on the other side, ablates it stage by stage, or measures third-party backbones (Table S7).

III. METHODS

A. Data

Probes are trained on the TUAB v3.0.1 *train* split [5, 16] (2,717 recordings, 2,076 patients; patient-stratified 90/10 train/validation, seed 7) and evaluated once on the official *eval* split ($n=276$, 253 patients, 150 normal, 126 abnormal; patient-ID intersection with train 0). No evaluation recording touches training, validation, calibration or the fixed-scale constants of the naive gain arm.

B. Models, adapters and classifiers

Three frozen public models (backbones) via braindecode 1.5.2: EEGPT (eegpt-pretrained), LaBraM (labram-pretrained) and BIOT (biot-pretrained-prest-16chs). Each consumes ESF through a fixed model-side adapter applied identically to every arm and cell: EEGPT, 19 channels at 250 Hz in 4 s windows, encoder tokens averaged to 512-d; LaBraM, 19 named channels at 200 Hz in its pretrained 15 s windows, patch tokens averaged to 200-d; BIOT, the 16 longitudinal bipolar derivations formed from the 19 referential channels at 200 Hz in 5 s windows, 256-d encoder embedding. EEGPT’s public checkpoint lacks its 19→19 channel projection; it is initialised from seed 7, frozen and released. Each backbone carries four probes on the pooled per-recording representation (mean, max, 95th percentile of ℓ_2 -normalised window vectors): three MLP probes ($3d \rightarrow 256 \rightarrow 1$, dropout 0.2, seeds 7/11/13, AdamW 10^{-3} , class-weighted BCE, 25 epochs, best validation AUROC) and one logistic probe; train-statistics standardisation, Platt scaling on validation. Windows are pooled in up to eight groups; the recording score is the mean group logit and the across-group variance is kept as a label-free monitoring statistic.

C. Pretraining provenance

Evaluation recordings must be unseen at every stage, including pretraining. EEGPT was pretrained on non-TUH data (PhysioNet motor imagery, the Tsinghua SSVEP benchmark, M3CV and SEED, per the authors’ repository). LaBraM’s pretraining includes TUAR, TUEP, TUSZ and TUSL but excludes TUAB and TUEV, recording-disjointly; patient-level overlap with TUAB is not reported and is a residual, unquantified risk for LaBraM only. For BIOT we use the checkpoint pretrained on Massachusetts General Hospital resting EEG alone; the six-dataset checkpoint, which includes TUAB train, is not used.

D. The contract

ESF v1.0: **(C1)** the 19 international 10–20 sites [17] in fixed order; **(C2)** 250 Hz by anti-aliased polyphase resampling; **(C3)** physical microvolts from the EDF header, then channel-wise robust z-score (median, IQR, persisted); **(C4)** common average reference over good channels; **(C5)** IIR notch at the regional mains and harmonics, and a 0.5 Hz high-pass; **(C6)** missing

channels are NaN with a quality flag, never imputed. Reference implementation: MIT.

E. Acquisition-shift axes

Every simulated difference is applied as an operator A_θ on the *raw* microvolt recording at its native rate, before any contract step (Fig. 1b; Table S6). The *canonicalized* arm then runs C1–C6; the *naive* arm runs the same pipeline with the one corrective step disabled, so the arms differ only in that step and share every downstream weight. One (difference, severity) setting is a *cell*; a model’s *envelope* is the set of cells in which the canonicalized arm stays within 0.01 AUROC of the model’s clean baseline. (A1) native rate $f_n \in \{128, 200, 250, 256, 512\}$ Hz by anti-aliased resampling; naive arm presents native samples as if 250 Hz. (A2) scalar gain $g \in \{0.5, 0.75, 1, 1.5, 2\}$; naive arm normalises with fixed train-set scale constants. (A3) 60 Hz mains and harmonics below native Nyquist, mean peak $a \in \{0, 5, 15, 30, 60\}$ μ V, per-electrode log-normal amplitude ($\sigma=0.3$) and random phase; naive arm has the notch off. (A4) montage *full19*, *legacy16* (T3, T4, T5, T6, Fz, Cz, Pz absent) or *bipolar18* re-projected to referential slots; naive arm averages over all 19 slots with zeros. (A5) white noise per channel at SNR $\in \{-5, 0, 5, 10, 20\}$ dB; no corrective stage exists, so the arms coincide. A sixth, measurement-only axis low-passes the raw recording at 15–100 Hz (§IV-B).

F. Recoverability framework

Let A_θ map a canonical signal to its native-device representation and K_θ be a canonicalization operator targeting $K_\theta \circ A_\theta = \text{id}$; f is the frozen probe and $\mathcal{A}(f; \mathcal{D})$ its AUROC. An axis is (f, ε) -recoverable iff some K_θ keeps $|\mathcal{A}(f \circ K_\theta \circ A_\theta) - \mathcal{A}(f)| \leq \varepsilon$ for every θ in the support.

Proposition 1 (Left-invertibility $\Rightarrow$ recoverability). *If A_θ admits a left-inverse within the contract, the axis is $(f, 0)$ -recoverable for every f .*

Proposition 2 (Sub-Nyquist irrecoverability). *If A_θ is an ideal anti-alias low-pass at $f_n/2$ followed by decimation with $f_n < 2f_{\max}$, and f depends on content in $(f_n/2, f_{\max}]$, no K_θ recovers f ’s clean AUROC.*

Theorem 1 (AUROC–TV bound, class-conditional). *For any bounded-score f , with $P|_{\pm}$ the class-conditional score distributions, $|\mathcal{A}(f; \mathcal{D}_0) - \mathcal{A}(f; \mathcal{D}_\theta)| \leq \text{TV}(P_0|_+, P_\theta|_+) + \text{TV}(P_0|_-, P_\theta|_-) \leq \sum_{\pm} \sqrt{\frac{1}{2} D_{\text{KL}}(P_\theta|_{\pm} \| P_0|_{\pm})}$.*

Proposition 1 applies to gain and mains. Proposition 2 is a statement about the classifier, not the axis. The marginal score distribution, which a deployed system can observe without labels, is *not* bounded this way (v1 stated a marginal form with $C(f) \leq 4$; that step was invalid). Proofs and a composability lemma: Appendix A.

G. Statistics

Canonicalized-vs-naive contrasts use DeLong’s test on matched recordings [18]; intervals are paired class-stratified

TABLE 1: **Envelope.** Change in AUROC from each model’s clean baseline on TUAB eval ($n=276$), per cell and arm, mean over three MLP seeds; shading is proportional to the loss, saturating at -0.5 . Bold: canonicalized-arm cells more than 0.01 below baseline, i.e. outside the envelope. A5 has no corrective stage, so its naive row is empty. Paired tests per contrast in Table S4.

		A1 rate (Hz)				A2 gain ($\times$)				A3 mains (μ V)				A4 montage		A5 SNR (dB)				
		128	200	256	512	0.5	0.75	1.5	2	5	15	30	60	leg16	bip18	-5	0	5	10	20
EEGPT	naive	-0.17	-0.02	-0.01	-0.17	-0.08	-0.04	-0.02	-0.02	-0.02	-0.10	-0.22	-0.30	-0.01	-0.02					
	canonicalized	0	0	0	0	0	0	0	0	0	0	0	0	-0.01	-0.02	-0.16	-0.11	-0.06	-0.04	-0.01
LaBraM	naive	-0.08	-0.05	-0.01	-0.29	-0.02	-0.02	-0.01	-0.01	-0.03	-0.09	-0.11	-0.21	-0.04	-0.02					
	canonicalized	-0.08	0	0	0	0	0	0	0	0	0	0	0	-0.04	-0.02	-0.20	-0.15	-0.12	-0.09	-0.05
BIOT	naive	-0.32	-0.05	0	-0.49	-0.06	-0.05	-0.02	-0.04	-0.24	-0.27	-0.25	-0.27	-0.03	-0.02					
	canonicalized	-0.23	0	0	0	0	0	0	0	0	0	0	0	-0.03	-0.02	-0.22	-0.11	-0.09	-0.06	-0.05

bootstraps (1,000 resamples); Benjamini–Hochberg at $q=0.05$ over the non-identity cells of each backbone [19]. Tables report the mean over the three MLP seeds; tests use the seed-7 probe.

IV. RESULTS

A. Baselines and per-axis results

On clean recordings the three models are equally good: AUROC 0.900 (EEGPT), 0.894 (LaBraM) and 0.898 (BIOT), each within 0.01 across classifier seeds. Table 1 gives every cell; Fig. 2a adds the seed spread; Tables S1–S4 add balanced accuracy, expected calibration error (ECE) and paired tests.

Sampling rate (A1). Resampling to 250 Hz undoes every rate for EEGPT, including 128 Hz, where the naive arm had lost 0.17. For BIOT and LaBraM it undoes 200, 256 and 512 Hz (the naive 512 Hz arm of BIOT is below chance) but not 128 Hz: BIOT falls from 0.90 to 0.67 and LaBraM from 0.89 to 0.82 whether or not the signal is resampled, and both push normal recordings toward abnormal. What is lost at 128 Hz is signal content above 64 Hz, and §IV-B shows which models use it.

Gain (A2). Per-recording robust scaling undoes any gain for every model. Without it, EEGPT and BIOT lose up to 0.08 at half gain and are already below baseline at unit gain, because fixed scaling constants are themselves a mis-normalization; LaBraM’s naive arm is largely tolerant.

Mains interference (A3). The notch undoes mains at every amplitude for every model. Without it, EEGPT loses 0.30 and BIOT 0.27 at 60 μ V, and from 15 μ V both call almost every recording abnormal; LaBraM loses 0.21 and, unlike the other two, drifts toward calling recordings normal (§IV-C).

Electrode montage (A4). Neither averaging over the present electrodes nor averaging over all 19 recovers the missing ones. The loss is small for EEGPT (0.01–0.02) and larger for LaBraM (0.04 on legacy16, with balanced accuracy near chance); it is in the acquisition, not the reference.

Broadband noise (A5). There is no corrective step, so the arms coincide and AUROC falls with signal-to-noise ratio for every model, most steeply for BIOT and LaBraM.

B. Bandwidth is a property of the model

To find out which models use the content that low-rate acquisition removes, we low-pass the raw recording at cutoffs from 15 to 100 Hz before the contract (Fig. 2b,c). EEGPT’s

ranking does not change at any cutoff, so nothing above 15 Hz enters its decision; its calibration does change below 30 Hz (balanced accuracy 0.81 $\rightarrow$ 0.65). BIOT and LaBraM lose accuracy as soon as the cutoff drops below 100 Hz, and their 64 Hz cells reproduce their 128 Hz cells exactly (0.68 and 0.82). Both were pretrained on 200 Hz input and tokenize the 64–100 Hz band; EEGPT works on time-domain patches and does not. For deployment the envelope therefore has two edges, one for ranking and a tighter one for calibrated decisions.

C. How individual recordings fail

Aggregate AUROC hides what happens to a given recording, so we follow each one across the 12 non-identity cells of the rate, gain and mains axes and sort it into one of seven categories (Fig. 3c; Table S5). The category that matters is *catastrophic shift*: a recording the model gets right on clean data and then, on at least one cell, gets wrong with probability 0.80 or more. Without the contract, this happens to every correctly classified normal recording for EEGPT (110/110) and BIOT (124/124), almost always toward a confident abnormal report; for LaBraM it happens to normals at 512 Hz (104/119) and, in the opposite direction, to abnormals under mains interference (90/103 at 60 μ V). Which way a model fails is therefore a property of the model and the difference together; for a screening tool the first direction costs review time [20], the second costs missed disease. With the contract, no confident flip remains for EEGPT, and none for BIOT or LaBraM outside the 128 Hz cell, where 100 of 124 and 79 of 119 correctly classified normals still flip. “Flips on some cell” is a maximum over 12 cells and describes this sweep, not a population rate.

D. Controls, monitor, bound

Could the classifiers have learned something other than the label? Classifiers trained on permuted labels (training and validation both permuted, ten repeats) score at chance for EEGPT (0.46 ± 0.08) and LaBraM (0.48 ± 0.10) and slightly above it for BIOT (0.58 ± 0.04). The BIOT residual is not leakage: with no labels at all, the distance of an evaluation recording from its ten nearest training recordings already separates the classes at 0.74 for BIOT (0.70 LaBraM, 0.51 EEGPT), and untrained random classifiers score at chance.

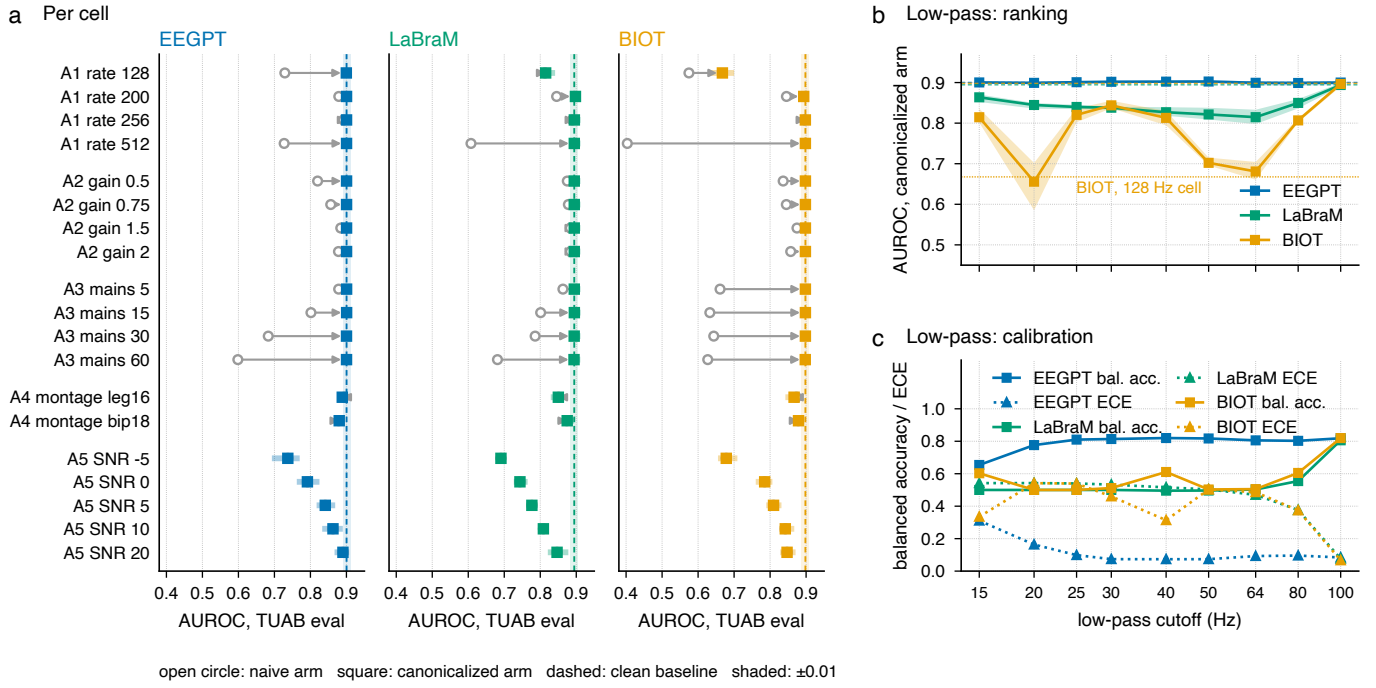


Fig. 2: **Every cell, and why the models disagree.** (a) AUROC of the naive arm (open circle) and the canonicalized arm (filled square) per model; the arrow is the recovery the stage delivers, the band the min–max over three probe seeds, the shaded strip the ± 0.01 envelope around the clean baseline. (b) AUROC of the canonicalized arm after a low-pass applied to the raw recording: EEGPT’s ranking is unchanged down to 15 Hz; BIOT’s and LaBraM’s 64 Hz cells reproduce their 128 Hz cells with no resampling. (c) Balanced accuracy and ECE over the same cutoffs.

Abnormal recordings are simply atypical in BIOT’s feature space. *Could a deployed system detect the failures without labels?* Neither the spread of a recording’s window scores nor its distance from a probability of 0.5 ranks the catastrophic recordings above the rest (median AUROC 0.54 over 11 naive cells); the failures look, to the model, like ordinary confident decisions. *Bound.* The class-conditional bound of Theorem 1 holds on all 40 measured arms for every model, and the label-free (marginal) version has no constant below 1 on any of them (Appendix A.4): the marginal shift a deployed system can see does not bound its loss.

E. A real legacy recorder

Six de-identified routine recordings from a Natus recorder in clinical use (125 Hz, 24–27 channels, 50 Hz mains, proprietary .e export; no reads available) were parsed, passed through the contract and scored locally. All three models call all six abnormal. The recorder’s 125 Hz sits inside EEGPT’s envelope and outside LaBraM’s and BIOT’s (Fig. 3b), so only EEGPT’s verdict could be tested against reads; whether the six are abnormal or merely shifted, we do not guess.

V. DISCUSSION

What the contract closes depends on the model. Gain and mains are undone for every model, as the theory predicts before any data. A low sampling rate is undone for EEGPT and not for the other two, and the low-pass sweep locates the reason in the 64–100 Hz band. A contract measured on

one model says nothing about another. Missing electrodes and broadband noise are undone for none; the practical response is to flag such recordings at intake by the size of the model’s output, since every model fails with high confidence and a rule based on borderline probabilities would miss exactly these failures.

Changing the contract. Because EEGPT uses nothing above 15 Hz, ESF v1.1 will add an explicit 0.5–50 Hz band-limit, as BRIDGE-EEG’s preprocessing does [13]; for a model of that bandwidth every native rate from 100 Hz upward is then on-contract by construction. For BIOT or LaBraM the same step would move the edge, not remove it: a step must be measured per model, not assumed.

Scope. One task on one public corpus, three frozen models, small classifiers, simulated rather than sampled recorders, and six unlabelled clinical recordings. LaBraM’s pretraining includes other TUH subcorpora (§III-C). No clinical claim is made; a labelled multi-recorder study is the next step (Appendix E).

VI. CONCLUSION

A preprocessing contract can be measured rather than assumed. Five simulated recorder differences, three frozen models, and each corrective step switched on and off give, per model, the differences the contract undoes, the step responsible and the calibration cost. Uncorrected, the models fail confidently and in model-specific directions; corrected, they fail only where the sweep says the contract cannot reach.

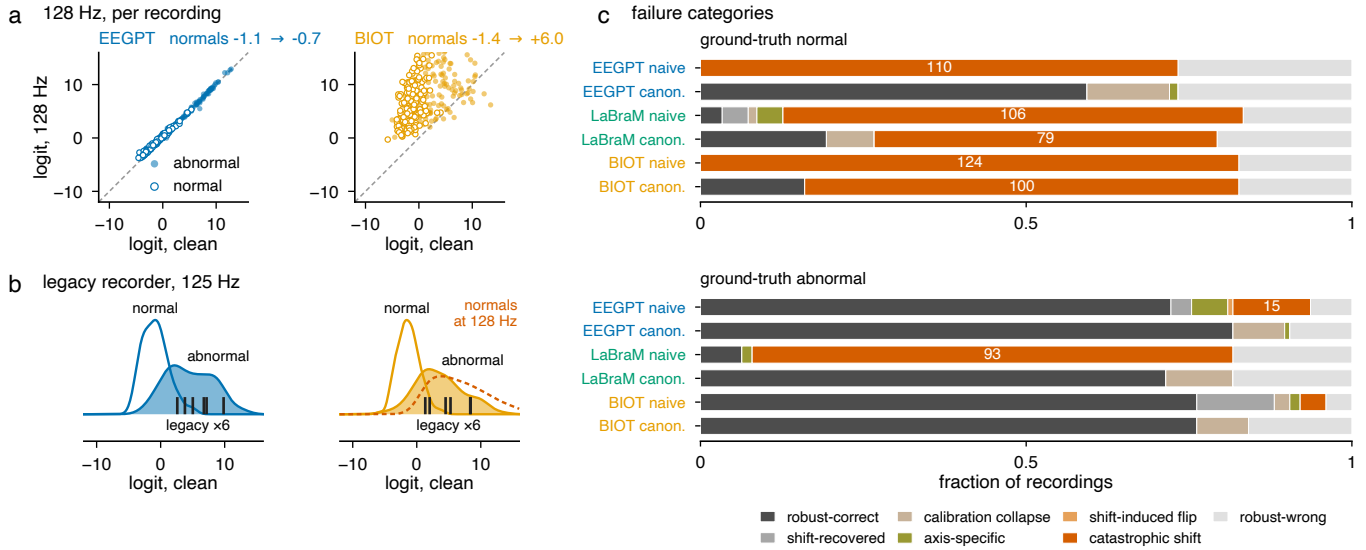


Fig. 3: **What the envelope means for a recording, a population and a recorder.** (a) Per-recording logit after 128 Hz acquisition and the full contract against the clean logit (seed-7 probes): EEGPT stays on the identity, BIOT lifts every normal recording by about 7 logits. (b) Clean eval logit distributions with the six legacy Natus recordings (125 Hz) as ticks; for BIOT the dashed curve is where its eval normals land at this rate. (c) Failure-category composition over the 12 non-identity cells of A1–A3, by model, arm and ground truth (counts in Table S5).

REPRODUCIBILITY

Code, operators, adapters, all twelve probes, per-cell predictions for every (model, probe, cell), tables and proofs: <https://github.com/h5371h/esf-recoverability> (MIT; tag v2.0.0-preprint-v2, archived as doi:10.5281/zenodo.22771925; the withdrawn v1 snapshot remains at v1.0.0-spm2026-submission). TUAB is obtained under the NEDC agreement [5]; the legacy recordings are clinical data, processed locally and not released. Checkpoint revisions, library versions and seeds: Appendix E. Every table and figure regenerates from the released per-recording predictions without a GPU; the feature and cell caches (about 12 GB) are not archived and regenerate from the released scripts in a few GPU-hours (Appendix E). *Cost.* On a 30-core CPU the full contract takes 0.64 s per 25-min recording (notch 0.28, EDF read 0.23, robust z 0.17, high-pass 0.16, CAR 0.02, resample < 0.01 s); the backbones take 12 (EEGPT), 1.6 (LaBraM) and 0.5 s (BIOT), or 113/10/3 s single-threaded. The contract is never the bottleneck.

VERSION NOTE (V2)

v2 corrects v1 in six respects, all found by the author after submission: v1 applied operators *after* canonicalization (its rate axis was a time-warp, its gain arm never composed forward and inverse, its mains amplitudes were in robust-SD units); its probe was trained on a split that overlapped eval; Theorem 1 was stated marginally; $C_{\max}$ was fitted, not validated; the conflict-of-interest statement was incomplete; the channel-projection adapter was unseeded. *Withdrawn:* the v1 losses at 128 and 512 Hz for EEGPT, the montage calibration recovery, the empirical support for Proposition 2. *Added:*

LaBraM, BIOT, seeds, a linear probe, bandwidth, envelope, controls, the legacy recorder. Details: Appendix E.

ACKNOWLEDGEMENTS

No external research funding. v1 compute: Microsoft for Startups Founders Hub; v2 compute: Lambda Cloud, paid by the author. The author thanks NEDC for TUAB and the EEGPT, LaBraM and BIOT authors for open weights. **Conflict of interest.** The author is the founder of Aposematium Private Limited, which develops the ENCEPHLIAN clinical-EEG software; the format described here shares that product’s name and the author has a commercial interest in its adoption. The reference implementation is MIT-licensed.

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